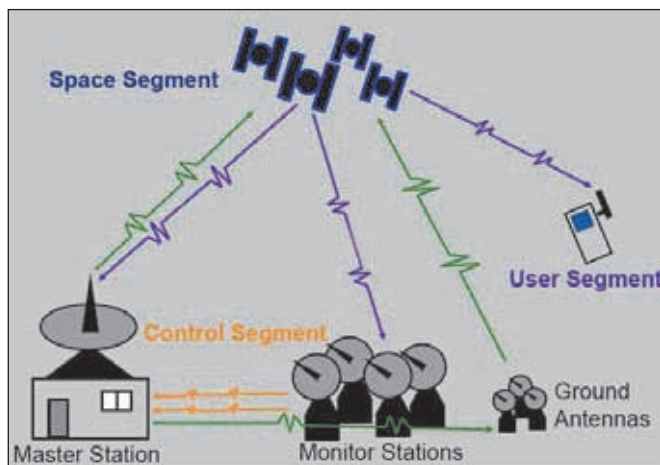


GPS receivers for precision weapon guidance. However, in 1983, Korean Air Lines Flight 007 was shot down for entering forbidden USSR airspace. This prompted then United States president Ronald Reagan to issue a directive that dictated that civilians would be allowed to use the GPS technology freely. Selective availability was ended in 2000, improving the precision of civilian GPS from about 100 m to about 20 m.

The GPS system originally consisted of 27 satellites that were launched between 1989 and 1993. Out of these 24 were fully functional while the rest are used only for storage purposes. However, as of March 2008, there were 31 active satellites, which further improves the usability and the accuracy of GPS.

The entire GPS system consists of three parts – the space segment (SS), control segment (CS) and the user segment (US). The SS consists of all the satellites that orbit the Earth as mentioned above. The trajectories of the satellites are such that at any given point in time, at any place, there will be at least four satellites which will be in visibility range with respect to the receivers.

The control segment consists of monitoring stations and a master control station. The monitoring stations are in Hawaii, Kwajalein, Ascension Island, Diego Garcia,



A representation of the segments in the GPS system